Adaptive Uncertainty Quantification for Maritime Classification under Cloud Cover in Satellite Imagery

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Abstract. Classifying vessels in satellite imagery is important for naval intelligence, search-and-rescue, and the monitoring of illegal activities. Performance, however, degrades under adverse meteorological conditions (e.g., clouds), which obscure discriminative features and increase ambiguity of class membership. Conventional deep learning methodologies fail to directly address this ambiguity, sometimes generating overly confident predictions for a specific erroneous class. To address this gap, we present WAVES (Weather-Aware Visual Estimation with Sets), a conformal prediction framework that makes uncertainty quality-aware. WAVES learns a temperature map that adjusts classifier logits as a function of predicted cloud coverage. A single global conformal threshold is then computed using the standard order statistic, preserving split-conformal marginal coverage while producing prediction sets that adapt across quality conditions. We evaluate WAVES using a publicly accessible dataset comprising various ship classes, which we augment with realistic synthetic clouds. Across three relevant classification models from the literature and compared against standard split–conformal prediction, WAVES typically maintains or improves coverage while reducing prediction–set size; across all three models, it reduces average set size by 6–7% on average (up to 15% in the best setting) while keeping coverage within ± 0.1 percentage points of the global split–conformal prediction baseline. These results indicate that quality-aware calibration combined with a global conformal threshold provides reliable, compact prediction sets for maritime surveillance under variable cloud cover, especially in operational scenarios where classification outcomes directly influence critical human or autonomous decisions.

1 Introduction

Automated classification of vessels in satellite imagery plays an important role in numerous maritime operations, including military intelligence, search-and-rescue missions, naval reconnaissance [7, 59], and surveillance of illegal maritime activities such as unreported fishing, piracy, and smuggling [17]. While the Automatic Identification System (AIS), a transponder-based vessel tracking system, supports vessel identification and location monitoring [25], it exhibits several limitations: AIS transponders may malfunction or be deliberately deactivated during covert naval operations or by vessels engaged in illegal activities [17, 25, 40, 63]. These limitations highlight the necessity for robust, alternative classification methods capable of operating independently of AIS data. As a result, maritime surveillance using satellite images has become important for monitoring shipping routes and maintaining maritime security [7, 10, 15, 17, 59].

However, meteorological conditions such as fog, haze, and cloud cover can substantially degrade image quality and occlude discriminative features, thereby increasing class ambiguity and degrading the performance of automated maritime vessel classification [1, 9, 17, 48, 61]. This is particularly problematic in fine-grained settings where classes (e.g., frigate, destroyer, corvettes, and cruisers) exhibit high inter-class similarity and share structural attributes [6, 62]. Synthetic Aperture Radar (SAR) can penetrate clouds but often lacks spatial detail and color cues needed for fine-grained classification [17]. Fig. 1 illustrates such fine-grained similarities in an annotated dataset; in this paper, however, our experiments consider image-level single-vessel classification (no detection or bounding-box inference), and Fig. 1 serves only as context for the visual challenges present in maritime scenes.

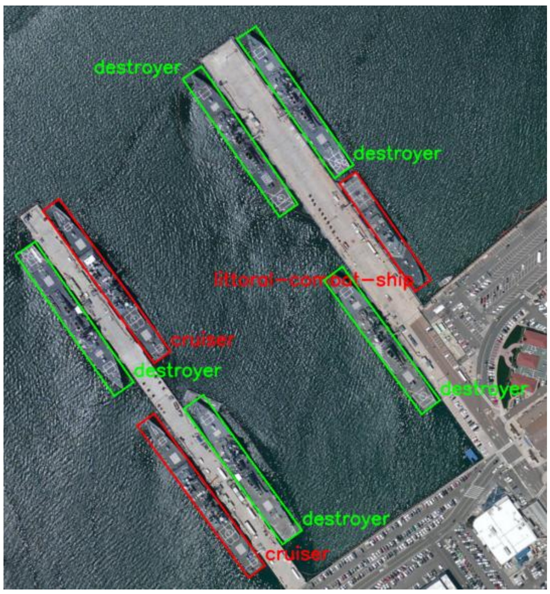

Figure 1. Example satellite image of military vessels from the FGSRCS dataset [48] with high inter-class similarity. The depicted ship classes include destroyer, cruiser, and littoral-combat-ship.

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Conventional deep learning methodologies, which are frequently used for marine vessel classification (e.g., in [10], [21], [27], and [60]), do not adequately address the potentially increased ambiguity of class membership due to meteorological conditions. These methodologies output a score vector over all classes that is routinely interpreted as a probability distribution. In practice, decisions are made by selecting a single most probable class (top-1) or by inspecting a top-k list. These procedures, however, do not provide statistical guarantees about whether the true class lies within the set of reported labels [2]. Moreover, although some models can express ambiguity via reduced class probabilities, they may still demonstrate overconfidence in incorrect predictions by attributing high probabilities to erroneous classes [4]. The implications of these errors can be severe: misclassifying a civilian vessel as a military threat, or vice versa, could lead to significant strategic errors or unintended engagements.

Conformal Prediction (CP) addresses this by constructing prediction sets that, under standard i.i.d. assumptions, contain the true class with user-specified coverage [2, 47]. In contrast to ad-hoc top-k selection, CP calibrates set size from data to target coverage and thus provides transparent uncertainty quantification. CP has already been successfully applied across diverse applications, including medical diagnostics [37], financial forecasting [3], and autonomous driving [31]. However, standard CP guarantees marginal coverage but does not, by itself, correct covariate–dependent miscalibration; its efficiency relies on calibration scores that accurately reflect uncertainty across samples. In practice, meteorological conditions can act as covariates that induce conditional miscalibration: under poor visibility, models may be overconfident on incorrect classes. If raw (visibility-agnostic) probabilities are used, the resulting nonconformity scores form a mixture over visibility regimes, which can inflate set sizes and waste efficiency. Prior work on adaptive CP shows that accounting for such covariates can improve efficiency and conditional behavior [13, 14, 38].

Motivated by this, we propose WAVES (*Weather-Aware Visual Estimation with Sets*), a quality-aware CP framework for maritime vessel classification from satellite imagery. WAVES learns a temperature map that adjusts classifier logits as a function of predicted image quality (here: cloud coverage as a proxy for visibility), so that probabilities are better aligned across cloud conditions. The same quality-aware calibration is then applied to compute a single global conformal threshold via the standard order statistic. This preserves split-conformal marginal coverage while producing prediction sets that adapt across quality conditions. In this work we instantiate image quality with cloud coverage; other factors (e.g., haze, fog, illumination) are discussed as extensions in future work.

Our main contributions are: (1) the first application of CP to satellite-based maritime vessel classification; (2) WAVES, a novel method that combines quality-conditioned temperature scaling with a single global conformal threshold, yielding compact prediction sets that adjust to cloud-induced visibility; and (3) a comprehensive experimental evaluation on a publicly accessible dataset [60] augmented with realistic synthetic clouds, using three relevant classification models from the literature. Compared to standard split-conformal prediction with a single global threshold, WAVES typically maintains or improves coverage while reducing prediction-set size, enabling reliable and compact uncertainty quantification for maritime surveillance under variable cloud cover.

The remainder of this paper is organized as follows: Section 2 reviews related work. Section 3 presents the proposed WAVES method in detail. Section 4 describes our evaluation setup and discusses the results. Section 5 concludes and outlines future work.

2 Related Work

2.1 Maritime Vessel Classification from Remote Sensing Imagery

Optical satellite imagery has become an important source of data for maritime vessel classification, leading to the development of various Fine-Grained Ship Classification (FGSC) datasets, such as ShipRSImageNet [61], FGSC-23 [60], FGSCR-42 [10], FGSCM-52 [27], and FGSRCS [48]. Zhang et al. [59] provide a comprehensive overview of over fifty ship detection and classification datasets (optical and SAR), whereas Kanjir et al. [22] review more than 100 related methodologies, highlighting developments driven by advancements in artificial intelligence and the growing availability of high-resolution annotated imagery.

Karus et al. [23] demonstrated the potential of vision transformers (on ship side views), while Huang et al. [21] combined Convolutional Neural Networks (CNNs) with transformers on satellite imagery to improve accuracy. Transfer learning has also been leveraged to bridge domain gaps from civilian to military vessels [58]. Beyond architectures, feature quality has been improved via Essential Feature Mining with iterative refinement [57], adaptive mid-level attention for subtle structural vessel details [55], and multi-scale contrastive learning with channel–spatial attention and a ResNet-50 backbone [11]. Robustness to resolution variation has been addressed through multigranularity feature enhancement strategies [49].

Moreover, contrastive and metric learning has been applied to maritime vessel classification. For example, region-based global–local contrastive learning to handle inter-class similarity and intra-class variations [29], hierarchical contrastive learning across optical and SAR modalities [7], and multiple-proxy deep metric learning with distribution optimization to maximize inter-class feature separability [53]. Hierarchical and semantic modeling further exploit label taxonomies [56], refined anchor labeling strategies and polarization functions [35], and textual guidance for visual feature extraction [34], while large vision–language models can be adapted efficiently via prompt tuning [27]. Additional approaches employed interpretable cognitive retrieval [52], spatial-invariant hashing with self-attention [18], and collaborative global–local knowledge distillation for imbalanced data [50].

Despite these advances, these existing methods yield score vectors, which do not provide calibrated probabilities that reflect true uncertainty with statistical guarantees, nor do they ensure that the true class lies within the reported top-1 or top-$k$labels [2]. Moreover, reduced softmax confidence is not a guarantee of ambiguity, and models can remain overconfident on erroneous classes [4]; a limitation that is relevant under weather-induced image quality degradations.

2.2 Impact of Meteorological Conditions on Remote Sensing Imagery

Optical satellite images, while capable of providing high-resolution details relevant for accurate vessel classification, are susceptible to quality variations caused by meteorological conditions, including fog, mist, haze, as well as clouds and their shadows [1, 9, 17, 22, 48, 61]. Moreover, datasets utilizing publicly available data, such as Google Earth, consist of images from different satellites with changing sensor characteristics, ranging from highly detailed to low-resolution images [59]. These factors substantially influence classification accuracy. However, existing publicly available datasets generally underrepresent images with adverse weather and environmental conditions, thereby constraining robustness in real-world maritime contexts [61]; for instance, only 11% of the images in [48] encompass more challenging conditions characterized by meteorological disturbances or other quality issues, while [17] entirely excludes cloudy images from their dataset. In contrast, datasets specifically designed for automotive applications often systematically include challenging weather conditions [5, 24, 46], an approach still largely lacking in maritime domains.

To directly mitigate image quality degradation, several deep learning-based image enhancement and restoration methods have been proposed. Techniques such as cloud removal and atmospheric correction have shown potential for improving optical satellite imagery. Lee et al. [28] demonstrated that CNNs effectively remove broader-scale environmental noise, however, they often struggle to preserve critical fine-grained features necessary for accurate maritime vessel classification. Xiang et al. [51] used complementary SAR images alongside optical images to mitigate thick cloud coverage; however, the availability of such paired multimodal data is typically limited. Huang et al. [20] proposed a Generative Adversarial Network (GAN)-based approach, which utilizes multiple images of the same geographical region captured under varying cloud conditions to reconstruct clear imagery. However, acquiring multiple temporally and meteorologically distinct observations of identical locations may prove challenging while vessels are in motion.

Beyond direct image restoration, several approaches have specifically aimed to enhance robustness to varying meteorological and environmental conditions in satellite imagery. Yang et al. [54] analyzed the influence of sea surface and extensive cloud coverage, demonstrating how self-similar regions can interfere with reliable vessel detection in large satellite images. Nelson et al. [39] tackled underrepresentation of illegal fishing vessels in varying maritime conditions using data augmentation techniques, using the POSEIDON framework [42]. Liu et al. [32] generated synthetic fog-occluded datasets to train models resilient to partial visibility conditions, achieving improvements in classification accuracy under increasing occlusion.

Further, multigranularity methods have been used to address variability in image resolution [49]. Chen et al. [7] applied hierarchical contrastive learning to integrate low-resolution and SAR imagery for coarse-grained classification, while high-resolution optical images were used for fine-grained distinctions. However, while these methods acknowledge image-quality issues, they typically do not provide explicit quantification of the uncertainty introduced by varying image conditions.

2.3 Uncertainty Quantification in Remote Sensing Imagery Analysis

Despite widespread application of artificial intelligence models, their adoption in high-risk and safety-critical applications remains limited, primarily due to a lack of reliable uncertainty quantification and transparency [12, 43]. CP offers a model-agnostic way to attach statistically valid uncertainty to any predictor: rather than returning a single most probable label (top-1) or a top-k list, CP returns a set of plausible labels, referred to as the prediction set, which is dynamically sized to meet a user-specified reliability level, referred to as the miscoverage $\alpha \in (0, 1)$. Conceptually, CP uses held-out calibration data to learn the empirical distribution of true-class scores (regardless of whether the model's top-1 prediction was correct) and, at test time, includes exactly those labels in the prediction set whose scores meet or exceed the calibrated cutoff, i.e., the conformal threshold learned from calibration. In particular, choosing a miscoverage level $\alpha \in (0, 1)$ targets a coverage guarantee of at least $1 - \alpha$, meaning the true label lies in the prediction set at least a $(1 - \alpha)$ fraction of the time. This guarantee holds with finite-sample uncertainty whose tightness depends on factors such as the size of the calibration set and the exchangeability of the data, i.e., calibration and test examples come from the same data-generating process and their order does not matter [2, 47].

In earth observation and remote sensing, CP has recently been employed in various applications. For land-use/land-cover mapping, CP has been used to produce pixel-wise prediction sets [45], and for hyperspectral image classification, spatial-aware variants leverage neighborhood information to aggregate nonconformity scores and improve set efficiency [33]. On aerial imagery, CP has demonstrated statistically reliable uncertainty even with limited labeled samples [41]. Beyond classification, CP has been explored for object detection in aerial/satellite imagery by quantifying detection and localization uncertainty [8], and for broader earth-observation pipelines as a practical, distribution-free uncertainty quantification tool [44, 26].

While CP guarantees marginal coverage, its efficiency, defined as achieving the target reliability with minimal prediction-set sizes, depends on how well calibration scores reflect uncertainty across operating conditions. In remote sensing, however, conditions are not homogeneous: meteorological factors (e.g., clouds, haze) can act as covariates that systematically shift confidence. That is, under poor visibility, models can be overconfident on the wrong class (e.g., assigning 80–90% to an incorrect label while giving only a few percent to the true one), whereas in clear scenes the pattern is typically reversed [4]. However, when raw, visibility-agnostic probabilities are pooled, the calibration scores mix clear and degraded regimes into a single empirical distribution. To satisfy the coverage target, the global quantile must then become more permissive to capture the hard, cloudy cases, thus effectively lowering the probability cutoff. This has two side effects: (i) in cloudy images, more additional (often incorrect) labels enter the set, and (ii) in clear images, the relaxed cutoff also admits extra labels that would otherwise be excluded. In both cases, prediction sets expand, reducing efficiency.

3 Proposed Method

3.1 Overview of the Proposed Method

The limitations identified in Section 2 highlight a gap in uncertainty quantification for maritime vessel classification under variable cloud cover: conventional classifiers do not fully adapt their confidence to visibility, and standard CP can be inefficient when calibration scores pool clear and cloud-degraded regimes. Inspired by adaptive conformal ideas for evolving operating conditions [38], we introduce WAVES (*Weather-Aware Visual Estimation with Sets*), a model-agnostic framework for adaptive, visibility-aware CP that explicitly conditions post-hoc calibration on predicted cloud coverage.

Aligned with this objective, WAVES first calibrates the classifier's outputs across visibility regimes and then applies a single global conformal threshold. Specifically, on a held-out temperature-calibration split it learns a piecewise temperature map that scales the model's logits (the pre-softmax class scores) as a function of predicted cloud coverage (our visibility proxy). The learned map is subsequently frozen and applied to calibration and test images, so that probabilities adapt to cloud-induced class ambiguity, i.e., models are made less overconfident under poor visibility while maintaining relatively high probabilities in clear conditions. From these calibrated probabilities, a single standard split-CP threshold is then computed on an additional conformal-calibration split using the nonconformity scores, and used thereafter for test-time set prediction.

Because WAVES operates as a post-hoc calibration layer followed by a single conformal threshold, it is model-agnostic, requires no retraining of the base classifier, and integrates seamlessly into operational pipelines.

3.2 Details of the Proposed Method

Inputs and prerequisites. We assume (i) a fixed, pre-trained *base classifier* that, for any image x, produces logits $z(x) \in \mathbb{R}^{|\mathcal{Y}|}$ and class probabilities $p(x) = \mathrm{softmax}(z(x))$ over a finite label set $\mathcal{Y}$; (ii) three disjoint splits: a *temperature-calibration* set $\mathcal{D}_{\mathrm{tc}}$ (used only to learn temperature parameters), a *conformal-calibration* set $\mathcal{D}_{\mathrm{cc}}$ (used only to compute the conformal threshold), and a *test* set $\mathcal{D}_{\mathrm{test}}$; and (iii) a *quality regressor* $r(\cdot)$ that maps an image to a scalar visibility proxy, here the predicted cloud-coverage score $\hat{q}(x) \in [0, 1]$ (0: clear, 1: fully cloudy). While other factors (haze, fog, illumination) may affect visibility, we deliberately focus on cloud cover and treat $\hat{q}(x)$ as the conditioning variable for calibration. The miscoverage $\alpha \in (0, 1)$ specifies the target reliability for CP, and a small number B of coverage bins defines the granularity of the temperature map learned on $\mathcal{D}_{\mathrm{tc}}$.

Temperature calibration. We learn a piecewise-constant temperature map $T(\hat{q})$ on $\mathcal{D}_{\mathrm{tc}}$ by partitioning the range of $\hat{q}$ into B quantile bins $\{I_b\}_{b=1}^{B}$ and selecting one temperature $T_b > 0$ per bin to minimize the negative log-likelihood on samples whose predicted coverage falls in that bin:

$$T(\hat{q}(x)) \;=\; T_b \quad \text{if} \quad \hat{q}(x) \in I_b. \tag{1}$$

Let $\tilde{p}(x; T)$ denote the temperature-scaled probability vector, obtained by dividing the logits by T and reapplying softmax:

$$\tilde{p}(x; T) \;=\; \mathrm{softmax}\!\left(\frac{z(x)}{T}\right). \tag{2}$$

The entry at index $y \in \mathcal{Y}$ of this vector is the calibrated probability $\tilde{p}_y(x; T)$ for class y.

For any single bin I_b, the temperature is chosen by minimizing the negative log-likelihood on samples whose predicted coverage falls into that bin:

$$T_b \;\in\; \arg\min_{t>0} \sum_{\substack{(x_j, y_j) \in \mathcal{D}_{\mathrm{tc}} \\ \hat{q}(x_j) \in I_b}} -\log \tilde{p}_{y_j}(x_j;\, t). \tag{3}$$

After fitting, we obtain the fixed, visibility-aware probability map

$$\tilde{p}_y(x) \;\equiv\; \tilde{p}_y\big(x;\, T(\hat{q}(x))\big), \tag{4}$$

which is then applied to the conformal-calibration and test sets.

Calibration of a conformal threshold. Given $\mathcal{D}_{\mathrm{cc}} = \{(x_i, y_i)\}_{i=1}^{n_{\mathrm{cc}}}$ (with $n_{\mathrm{cc}} = |\mathcal{D}_{\mathrm{cc}}|$), we apply the learned temperature map to obtain calibrated probabilities $\tilde{p}(x_i)$. In split-CP, a nonconformity score quantifies how atypical a labelled example appears under the model, i.e., larger scores indicate lower model support for the true labels. The nonconformity scores are calculated as follows:

$$s_i \;=\; 1 - \tilde{p}_{y_i}(x_i), \qquad i = 1, \ldots, n_{\mathrm{cc}}. \tag{5}$$

For example, if a calibrated probability vector for a three-class problem is $\tilde{p}(x) = \big[0.70,\ 0.20,\ 0.10\big]$ and $y = 1$, then the nonconformity score is $s = 1 - \tilde{p}_1(x) = 1 - 0.70 = 0.30$.

For a $\alpha \in (0, 1)$, the conformal threshold is the k-th order statistic of $\{s_i\}$,

$$\tau \;=\; s_{(k)}, \qquad k \;=\; \Big\lceil (n_{\mathrm{cc}} + 1)\,(1 - \alpha) \Big\rceil, \tag{6}$$

where $s_{(1)} \le s_{(2)} \le \cdots \le s_{(n_{\mathrm{cc}})}$ denotes the sorted scores. This single τ, computed after visibility-aware temperature calibration, is held fixed and used for all test instances.

Inference Phase. Given a new input x_{test}, we predict its cloud coverage $\hat{q}(x_{\mathrm{test}})$, apply the learned temperature map, and compute calibrated probabilities:

$$\tilde{p}(x_{\mathrm{test}}) \;=\; \mathrm{softmax}\!\left(\frac{z(x_{\mathrm{test}})}{T(\hat{q}(x_{\mathrm{test}}))}\right). \tag{7}$$

We then form the prediction set:

$$\hat{C}(x_{\mathrm{test}}) \;=\; \big\{\, y \in \mathcal{Y} : \tilde{p}_y(x_{\mathrm{test}}) \;\ge\; 1 - \tau \,\big\}, \tag{8}$$

Since the true label is unknown at inference, we evaluate scores for all classes and include those whose calibrated probability exceeds τ.

Under exchangeability, split-CP guarantees marginal coverage at least $1-\alpha$. Because probabilities are calibrated as a function of cloud coverage before conformalization, the single global τ is learned from scores that better capture visibility-dependent uncertainty, yielding more compact sets in clear conditions and appropriately larger (but calibrated) sets under heavy cloud.

4 Evaluation & Discussion

4.1 Dataset and Preprocessing

In our evaluation, we utilize the FGSC-23 dataset originally introduced by Zhang et al. (2020) [60], comprising 4,080 optical satellite images categorized into 23 fine-grained ship classes, including both military (e.g., aircraft carrier, destroyer, submarine) and civilian vessels (e.g., container ship, oil tanker, passenger ship). Images were primarily sourced from Google Earth and the Chinese Gaofen-1 satellite. The dataset features visually diverse images captured under varying illumination conditions, backgrounds, and resolutions. Notably, certain classes such as destroyers, cruisers, and frigates exhibit high inter-class similarity, making them particularly susceptible to class ambiguity under challenging meteorological conditions. The class distribution is notably imbalanced, reflecting realistic scenarios where certain ship classes occur more frequently, e.g. destroyer 542 times and medical ships only 27 times.

The original FGSC-23 dataset did not contain images with cloud-induced occlusion of vessel features. To systematically evaluate our WAVES method under such degraded imaging scenarios, we augmented the original dataset by applying synthetic cloud coverage using the method proposed by Czerkawski et al. (2023) [9]. Specifically, we introduced three levels of cloud coverage severity, categorized as *Mild* (on 55% of images), *Moderate* (on 35% of images), and *Severe* (on 10% of images). Each category differs primarily in cloud opacity, shadow intensity, and number of cloud layers. These synthetic clouds were generated by applying Perlin noise patterns combined with adjustable transparency masks, Gaussian blurring, and additional features including cloud color adaptation and channel-specific magnitude adjustments [9]. Due to the synthetic generation approach, a ground-truth cloud coverage score between 0 (clear) and 1 (fully cloudy) was directly available. In practical scenarios without synthetic augmentation, this score could alternatively be obtained

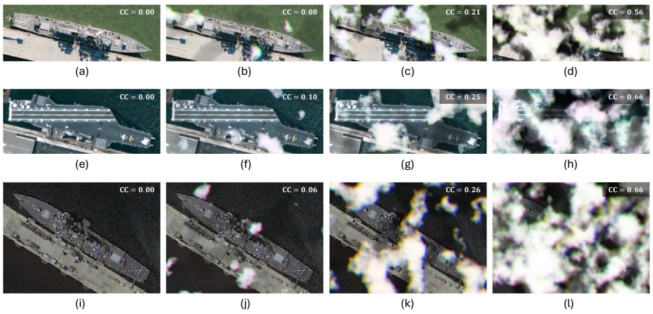

Figure 2. Synthetic cloud augmentation at four severities: (a,e,i) Raw image, (b,f,j) Mild, (c,g,k) Moderate, (d,h,l) Severe. Top row: Destroyer (class 12); middle row: Aircraft carrier (class 0); bottom row: Destroyer (class 12). For each image, the cloud coverage (CC) value is displayed in the upper-right corner.

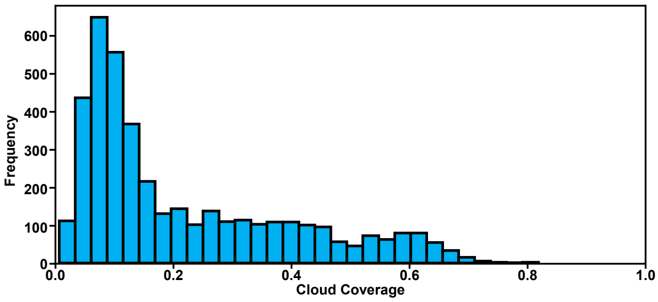

Figure 3. Distribution of synthetic cloud coverage scores in the modified FGSC-23 dataset. Most images exhibit mild to moderate cloud coverage, while a smaller proportion reflects severe coverage.

through human annotation or automated cloud-detection methods, e.g., those discussed in [30].

Visual examples showing different levels of synthetic cloud severity (*Mild*, *Moderate*, *Severe*) and their effects on image visibility are provided in Fig. 2. Additionally, the distribution of synthetic cloud coverage scores across the augmented dataset is shown in Fig. 3.

4.2 Experimental Setup

To evaluate WAVES, we conducted experiments using three relevant deep neural network architectures commonly employed in fine-grained visual classification tasks: ResNet-50 [16], ConvNeXt-Tiny [36], and DenseNet-121 [19]. All models were initialized with pretrained weights obtained from the ImageNet dataset and subsequently fine-tuned on our augmented FGSC-23 training subset, comprising 2,448 images (60%). Each image was resized and padded to a resolution of 384×384, and input normalization was performed using standard ImageNet mean and standard deviation values. The fine-tuning process employed the Adam optimizer with a learning rate of 3×10^{-4}, batch size of 8, a warm-up phase of head-only training for 3 epochs at a head learning rate of 1×10^{-3}, and a maximum training duration of 15 epochs. To address the inherent class imbalance in the dataset, we utilized a weighted random sampling approach during training. Our primary objective was not to achieve state-of-the-art classification accuracy or conduct extensive hyperparameter optimization, but rather to establish a representative set of baseline classifiers and clearly illustrate the increased classification difficulty introduced by realistic cloud coverage conditions in maritime vessel classification.

The dataset was split as follows: 2,448 images for training (60%), 407 for validation (10%), 816 for conformal calibration (20%), and 409 for testing (10%). The validation split was used for hyperparameter tuning and temperature calibration, while the conformal-calibration split was used only to compute the conformal threshold. Model and CP performance were then evaluated on the independent test split. All splits were generated via stratified sampling to preserve the original class distributions; class proportions per split are shown in Fig. 4.

We trained an auxiliary regression model based on the ResNet-18 architecture [16] to predict cloud-coverage scores (0: clear, 1: fully cloudy) from input images. This model achieved a Mean Squared Error (MSE) of approximately 9×10^{-4} on the test set, indicating accurate estimation of cloud coverage. To avoid target leakage, we used the predicted coverage scores $\hat{q}(x)$ for all downstream steps.

For CP, we evaluated miscoverage levels $\alpha \in [0.01, 0.05]$ in steps of 0.01. In WAVES, temperature calibration on $\mathcal{D}_{\mathrm{tc}}$ partitions predicted cloud coverage into B bins, and we assess multiple settings $B \in \{1, 2, 3, 4\}$; within each bin, a temperature is selected by grid search over $[0.5, 5.0]$ with step 0.1. For comparison, we include a *Global CP* baseline—standard split-CP without temperature scaling—that computes a single threshold directly from the raw probabilities.

4.3 Evaluation Results & Discussion

First, we assessed the baseline classifiers trained on the augmented FGSC-23 dataset. The overall accuracy achieved by the classifiers on the test set was 62.8% (ResNet-50), 64.6% (ConvNeXt-Tiny), and 59.9% (DenseNet-121). In preliminary experiments using the original FGSC-23 dataset without synthetic cloud augmentation, the examined classifiers achieved classification accuracies on the test set between 75.1% (ResNet-50) and 84.4% (DenseNet-121).

Fig. 5 visualizes how prediction confidence and accuracy vary with cloud cover on the test set averaged over all examined models. The blue curve (median raw top-1 probability) with its interquartile range (IQR), i.e., the range between the 25th and the 75th percentile, remains comparatively high even as cloud cover increases, while the green curve (top-1 accuracy) steadily declines and approaches zero in the highest-coverage bin. This gap indicates overconfidence under poor visibility: the models often assigns high probability to an incorrect class. This trend aligns with our aggregate finding that misclassified images exhibit substantially higher cloud coverage (mean ≈ 0.27) than correctly classified images (mean ≈ 0.18). These results indicate that visibility is a dominant driver of both accuracy loss and raw-probability overconfidence in maritime vessel classification.

Next, we examined the global CP baseline, reporting coverage and prediction–set sizes in Table 1. At a strict target of 99% ($\alpha = 0.01$), global CP produced large sets, with test-time average sizes ranging from 9.817 (ConvNeXt-Tiny) up to 13.171 (ResNet-50). At more relaxed targets (e.g., 95%, $\alpha = 0.05$), sets became smaller—down to 4.966 (ResNet-50)—indicating that several classes become hard to distinguish under a combination of cloud cover and inherent

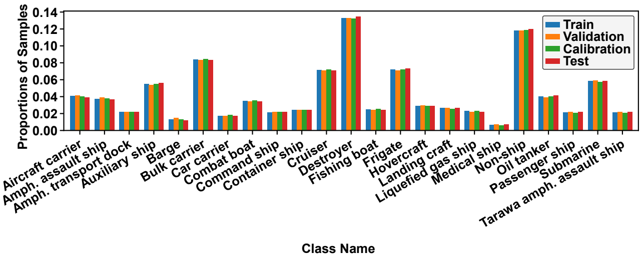

Figure 4. Relative class distributions of the training, validation, calibration, and test splits of the FGSC-23 dataset after stratified sampling.

Table 1. Global CP vs. WAVES on the test set. Each cell shows coverage and average set size (± standard deviation). Bold indicates the highest coverage and the smallest set size per row (miscoverage α).

Model	α	Global CP	WAVES (B=1)	WAVES (B=2)	WAVES (B=3)	WAVES (B=4)
ResNet-50	0.01	**0.993** 13.171 ± 7.649	**0.993** 12.286 ± 7.177	**0.993** 12.068 ± 7.020	**0.993** 12.198 ± 7.042	0.990 **11.208** ± 6.824
	0.02	0.978 8.936 ± 6.646	**0.985** 8.218 ± 5.774	0.983 7.716 ± 5.437	0.983 **7.702** ± 5.403	0.980 8.115 ± 5.616
	0.03	**0.973** 7.051 ± 5.698	0.971 6.699 ± 4.958	0.971 **6.616** ± 4.771	**0.973** 6.633 ± 4.710	0.968 6.692 ± 4.780
	0.04	0.946 5.386 ± 4.576	**0.954** 5.098 ± 3.830	0.951 4.936 ± 3.634	0.949 **4.778** ± 3.480	0.951 4.875 ± 3.552
	0.05	**0.944** 4.966 ± 4.298	**0.944** 4.555 ± 3.422	0.941 4.465 ± 3.292	0.939 **4.340** ± 3.151	**0.944** 4.575 ± 3.349
ConvNeXt-Tiny	0.01	0.983 9.817 ± 6.065	0.983 9.308 ± 5.195	0.983 **8.912** ± 5.501	0.983 9.059 ± 5.074	**0.988** 9.306 ± 5.574
	0.02	**0.971** 7.083 ± 4.961	**0.971** **6.653** ± 4.058	0.968 7.164 ± 4.598	**0.971** 6.665 ± 4.042	**0.971** 7.222 ± 4.565
	0.03	0.963 6.205 ± 4.472	**0.968** 6.029 ± 3.704	0.963 6.100 ± 3.918	0.963 6.046 ± 3.701	0.963 **5.944** ± 3.770
	0.04	0.961 5.880 ± 4.307	**0.963** 5.501 ± 3.398	**0.963** 5.210 ± 3.321	**0.963** **5.186** ± 3.227	**0.963** 5.333 ± 3.397
	0.05	0.956 5.308 ± 3.937	0.958 4.826 ± 2.985	0.958 **4.623** ± 2.978	0.958 4.780 ± 2.980	**0.961** 4.658 ± 2.991
DenseNet-121	0.01	0.976 12.029 ± 7.140	0.976 11.995 ± 6.856	**0.983** 12.528 ± 7.170	0.978 12.098 ± 7.170	0.980 **11.704** ± 7.172
	0.02	**0.976** 10.496 ± 6.889	**0.976** **9.756** ± 6.311	0.973 10.548 ± 6.876	0.973 10.496 ± 6.888	0.971 9.946 ± 6.837
	0.03	**0.968** 8.853 ± 6.410	0.961 8.411 ± 5.929	0.961 **8.389** ± 6.115	0.963 8.682 ± 6.334	0.961 8.814 ± 6.489
	0.04	**0.961** 7.939 ± 6.075	0.951 7.435 ± 5.419	0.954 **7.200** ± 5.470	0.956 7.203 ± 5.608	0.956 7.306 ± 5.770
	0.05	0.946 6.726 ± 5.453	**0.949** 6.408 ± 4.756	0.936 6.044 ± 4.653	0.934 **6.024** ± 4.737	0.934 6.130 ± 4.948

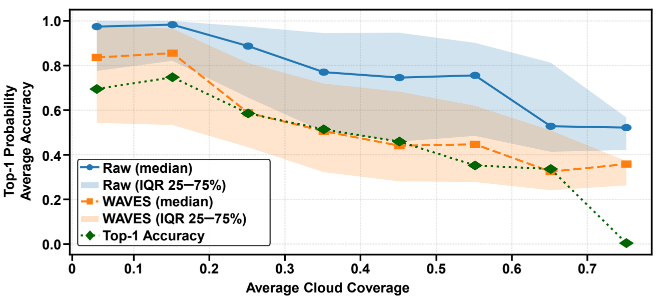

Figure 5. Top-1 probability and top-1 accuracy versus ground-truth cloud coverage on the test set. Shown are raw top-1 probability (median) with its IQR (25–75%) band, WAVES-calibrated top-1 probability (median) with its IQR (25–75%) band, and the top-1 accuracy curve.

inter-class similarity. In many cases the empirical test coverage falls short of the nominal target; this is expected because the split-CP guarantee is marginal and comes with finite-sample uncertainty that depends strongly on calibration-set size.

Fig. 5 aggregates the temperature-scaling effect of WAVES over all examined models and bin settings. The orange curve (WAVES top-1 probability, median) lies substantially below the raw top-1 probability across most cloud coverage bins, while top-1 accuracy remains identical, i.e., temperature scaling preserves the ranking of class probabilities. The largest reduction in median confidence occurs for cloud coverage between roughly 0.25 and 0.55, where the WAVES curve aligns most closely with observed accuracy, indicating that probabilities better reflect true uncertainty in this regime (reduced overconfidence). Under extreme occlusion (e.g., coverage $\approx$ 0.75), accuracy approaches zero; WAVES reduces median confidence compared to raw data, yet an offset remains between confidence and accuracy, underscoring residual uncertainty that cannot be entirely mitigated when the visual signal is significantly degraded.

In comparison to the global CP baseline, Table 1 reports test coverage and mean set size for WAVES across four bin settings $B \in \{1, 2, 3, 4\}$. Note that B=1 corresponds to a single bin (i.e., no cloud-dependent segmentation) and thus applies an overall probabilistic adjustment without conditioning on cloud coverage. Relative to the respective global CP values, in one third of all model–α–bin combinations WAVES simultaneously increases coverage and reduces set size (although this occurs in only 10% of DenseNet-121 cases). In an additional $\approx$ 27% of cases, WAVES maintains identical coverage while reducing set size. Hence, in the majority of settings WAVES outperforms the global CP baseline. In a further 30% of cases there is a trade-off: coverage decreases on average by 0.62% while the set size reduces on average by 7.45% (relative to the global CP set size). In the remaining 10%, we observe mixed outcomes: either set size is reduced while coverage improves (3.33%), or coverage decreases while set size is maintained (1.67%), or both coverage decreases and set size increases (5%).

Aggregating by the number of bins across all models and α-values, the largest average coverage increase occurs with B=1 (+0.05% on average), while still reducing mean set size by 5.76% on average. The largest average set size reduction arises with B=3 (−7.33% on average) at the expense of a modest average coverage decrease (−0.11%). The best overall trade-offs are observed with B=2 and B=5, both achieving an average set size reduction of 6.72% together with a small average coverage decrease of 0.09%. These patterns suggest that binning is particularly beneficial when reducing set size is prioritized and a small coverage decrease is acceptable.

Averaging per model over all α and B, WAVES performs best on ResNet-50 with an average set size reduction of 8.88% while increasing average coverage by 0.10%. For ConvNeXt-Tiny, WAVES achieves a 6.53% average reduction in set size and a 0.13% average increase in coverage. DenseNet-121 is the only model showing an overall trade-off: average set size reduces by 4.49% while average coverage decreases by 0.41%. The largest individual average set size reduction is obtained with ResNet-50 at $\alpha{=}0.01$ and $B{=}4$ (-14.90%), accompanied by a coverage decrease of 0.3%. The best individual trade-off (equal weight on coverage increase and set size decrease) occurs with ResNet-50 at $\alpha{=}0.02$ and $B{=}3$, where coverage increases by 0.5% and set size decreases by 13.81%.

Overall, these evaluation results underscore that WAVES effectively balances prediction-set informativeness with reliable true-class coverage, offering practical gains over global CP for maritime surveillance under variable cloud cover.

5 Conclusions & Future Work

In this paper we introduced WAVES, a model-agnostic conformal prediction framework for maritime vessel classification that explicitly addresses uncertainty quantification under variable cloud cover. To mitigate systematic overconfidence under poor visibility, WAVES first learns a visibility-conditioned temperature map on a held-out split to recalibrate logits, and then applies a single global split-conformal threshold on the calibrated scores. This preserves finite-sample marginal validity while aligning probabilities across cloud regimes and improving efficiency. On the FGSC-23 dataset with realistic synthetic clouds and three relevant classifiers, WAVES typically maintained or improved coverage while reducing prediction-set size relative to a standard global split-conformal prediction baseline. Aggregated across the examined models, WAVES reduced mean set size by approximately 6–7% on average (up to 15% in the best setting) while keeping coverage within ± 0.1 percentage points of global conformal prediction. Qualitatively, temperature scaling lowers raw top-1 probabilities in cloudy conditions where models tend to be overconfident, bringing them into better agreement with observed accuracy.

While our evaluation utilized synthetic cloud augmentation, validating WAVES on naturally occurring meteorological disturbances in operational satellite imagery would strengthen its practical applicability. Beyond cloud coverage, future research could explore incorporating additional quality factors such as haze, fog, mist, as well as different illumination conditions and sensor characteristics into the quality assessment. Moreover, we used a small number of visibility bins for temperature mapping; learning smoother (e.g., monotone or spline-based) temperature functions, incorporating spatial context, or jointly modeling uncertainty in the quality regressor may further improve conditional behavior. Finally, developing theoretical guarantees for the adaptive conformal prediction under distribution shift caused by varying image quality would strengthen the method's theoretical foundation and potentially extend its applicability to other domains facing similar quality-dependent uncertainty challenges.

Code & Data Availability Statement

The Python code for WAVES as well as the global conformal prediction method, the augmented dataset, the trained classification models, the auxiliary regression model, and the evaluation results are publicly available via https://github.com/GianMan89/uncertainty-aware-ship-classification.

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